Jiageng Wang

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Education

Nanyang Technological University — Singapore
M.Sc. in Analytics — *Aug. 2025 – Present*

Sun Yat-sen University — Shenzhen, China
B.Eng. in Aerospace Engineering — *Sep. 2021 – Jun. 2025*

Research Experience

Institute for Infocomm Research (I^{2}R), A*STAR — Singapore
Research Intern — *Oct. 2025 – Present*

SizingAgent: Physics-Grounded LLM Reasoning for Analog Sizing — *NeurIPS 2026 (1st author, under review)*

- Designed a training-free **LLM agent framework** with three modules (Physics Initializer, Physics-Guided Reasoner, Sizing Verifier) and two stabilising strategies (multi-start initialisation, sensitivity-guided search), applied to analog sizing as a high-stakes expert-domain task; each proposed action is verified against a precomputed sensitivity matrix before any external tool call.
- Constructed a **28-task LLM-evaluation benchmark** (4–17 parameters per task, four spec families) under a schema-driven task definition and a unified tool-execution wrapper; verified through 256-sample quasi-random feasibility scans that random search achieves near-zero success, ensuring the benchmark cannot be solved by brute-force sampling.
- Achieved **76.4% Pass@1** with Gemini 2.5 Flash (+**35.9 pp** over the strongest LLM baseline EEsizer) and **89.8%** with GPT-5.5; lifted open-weight Qwen3-32B from **9.7% to 57.0%** (+47.3 pp) with no fine-tuning, demonstrating direct **on-premises LLM** portability.
- Ablation isolates the LLM-derived initialisation step as the dominant contributor (−38.3 pp under random sampling), with each remaining agent module providing orthogonal gains on different task tiers; full code, benchmark, and four LLM-baseline re-implementations to be open-sourced under Apache 2.0.

AnalogAgent: Self-Improving Multi-Agent LLM for Analog Design — *KDD 2026 (2nd author, **Oral**)* [arXiv]

- Co-developed a training-free **multi-agent LLM framework** with three specialised agents (Code Generator, Design Optimizer, Knowledge Curator) coordinated through a **Self-Evolving Memory** module that converts tool-execution feedback into reusable agent prompts for cross-task transfer.
- Led **small-LLM adaptation** on Qwen3 (1.7B–14B) for **on-premises deployment under NDA constraints**, lifting Pass@1 from **28.2 to 62.3** (4B), **23.3 to 72.1** (8B), and **35.3 to 76.7** (14B), showing the agent framework can substitute for model scale.
- Achieved **97.4% Pass@1** with GPT-5 and **92.0%** with Gemini-2.5-Flash on a 30-task expert-domain benchmark, **12.5 pts** above the strongest baseline on Hard tasks; removing the memory module dropped Hard-task Pass@1 from **89.8% to 43.7%**.

CABAgent: Layout-Aware LLM Agent for Analog Benchmark Generation — *ICCAD 2026 (co-1st author, under review)*

- **IEEE SSCS Code-a-Chip Winner at VLSI 2026**; selected as one of **3 winning teams among all submissions** for an open-source, reproducible analog design automation notebook/poster.
- Extended the multi-agent framework to layout-aware automation: a self-evolving agent loop (generation, tool-execution validation, reflection, knowledge curation) produces SKY130-compatible netlists and expands each seed into benchmark packages with auto layout, verification, and post-layout evaluation under a single agent driver.
- Achieved **98.3% Pass@1** (**100% Pass@5**) on analog circuit generation with Gemini; produced **20 topologies and 1,000 benchmark packages within 10 hours** on a single workstation, establishing an end-to-end agent-driven benchmark pipeline.

Projects

Causal RL Portfolio System for A-Share Tech Stocks — *Bloomberg Code Crunch, Oct. 2025*

- Built an **RL portfolio framework** for 50 A-share tech stocks, integrating causal regression for supply-chain shock transmission, multi-agent credit assignment for joint multi-asset actions, risk-sensitive distributional policy optimization, and a sparse graph-attention module that scales to large stock universes.
- Implemented an asynchronous actor–learner training pipeline (V-trace, asymmetric policy clipping); completed all 9 system modules and validated through walk-forward backtests on out-of-sample windows.

Internship

Keyone AI Ltd. — Shenzhen, China
AIGC Algorithm Engineering Intern — *Apr. 2025 – Aug. 2025*

- Led full-stack development of a production-grade **Photoshop plugin** delivering an AI-powered background removal tool to enterprise clients (React front end, UXP/JavaScript back end).
- Deployed the **12B-parameter FLUX.1** image generation model on Linux servers; mitigated GPU memory bottlenecks via inference-side optimization and built a local model-artifact migration pipeline for offline deployment under restricted network environments.
- Directed the full-lifecycle upgrade of the **DBNet++** text detection model, including C++/CUDA operator modifications and framework refactoring.

Technical Skills

LLM & Agents: multi-agent orchestration, tool-use agents, prompt optimization, self-evolving memory, LLM evaluation, benchmark construction, small-LLM adaptation
ML Systems: PyTorch, Hugging Face Transformers/Diffusers, vLLM, Ollama, FastAPI, Linux, Git, CUDA/C++
Programming: Python, C/C++, JavaScript/React/UXP, MATLAB
Languages: Mandarin (Native), English (Fluent, IELTS 7.0)